A REPORT OF FOUR WEEK TRAINING

at

GURU NANAK DEV ENGINEERING COLLEGE LUDHIANA

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD

OF THE DEGREE OF

BACHELOR OF TECHNOLOGY

Information Technology

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SUBMITTED BY:

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DEPARTMENT OF INFORMATION TECHNOLOGY

GURU NANAK DEV ENGINEERING COLLEGE LUDHIANA

(An Autonomous College Under UGC ACT)

TRAINING CERTIFICATE

TO WHOM IT MAY CONCERN

This is to certify that **Mr. Prabhjot Singh**, Roll No./URN **2514657,** a student of B.Tech. Information Technology (IT), Guru Nanak Dev Engineering College (GNDEC), Ludhiana, has successfully completed the TR-101: Tech Foundations – Linux and Web Basics training conducted by the Department of Information Technology from June 2026 to July 2026.

During the training period, the student successfully completed practical exercises, assignments, and projects in Linux Operating System and Web Development (HTML & CSS) under the guidance and supervision of the faculty members of the Department of Information Technology.

During the training, the student demonstrated sincerity, discipline, enthusiasm, and a willingness to learn fundamental computing concepts and technologies. His overall performance during the training period was found to be satisfactory.

We wish him/her every success in future academic and professional endeavors.

Date: _____

Signature of the Student

Training Coordinator

Hea

d of Department

Department of Information Technology

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This is to certify that **Mr. Pulkit Bansal**, Roll No./URN **2514664,** a student of B.Tech. Information Technology (IT), Guru Nanak Dev Engineering College (GNDEC), Ludhiana, has successfully completed the TR-101: Tech Foundations – Linux and Web Basics training conducted by the Department of Information Technology from June 2026 to July 2026.

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Date: _____

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Training Coordinator

Hea

d of Department

Department of Information Technology

TRAINING CERTIFICATE

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This is to certify that **Mr. Sehajpal Singh Deol**, Roll No./URN **2514687,** a student of B.Tech. Information Technology (IT), Guru Nanak Dev Engineering College (GNDEC), Ludhiana, has successfully completed the TR-101: Tech Foundations – Linux and Web Basics training conducted by the Department of Information Technology from June 2026 to July 2026.

During the training period, the student successfully completed practical exercises, assignments, and projects in Linux Operating System and Web Development (HTML & CSS) under the guidance and supervision of the faculty members of the Department of Information Technology.

During the training, the student demonstrated sincerity, discipline, enthusiasm, and a willingness to learn fundamental computing concepts and technologies. His overall performance during the training period was found to be satisfactory.

We wish him/her every success in future academic and professional endeavors.

Date: _____

Signature of the Student

Training Coordinator

Hea

d of Department

Department of Information Technology

GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA

CANDIDATE'S DECLARATION

We, **Prabhjot Singh(2514657), Pulkit Bansal(2514664), Sehajpal Singh Deol(2514687)** hereby declare that we have successfully undertaken the TR-101: Tech Foundations –Linux and Web Basics training organized by the Department of Information Technology, Guru Nanak Dev Engineering College, Ludhiana during the period from June to July 2026 in partial fulfillment of the requirements of the B.Tech programme.

We further declare that the work presented in this training report, submitted to the Department of Information Technology, Guru Nanak Dev Engineering College, Ludhiana, is an authentic record of the training undertaken by us. The report has been prepared on the basis of the practical work, assignments and projects completed during the training, and is an authentic record of training work

Signature of the Student

The four week industrial training Viva–Voce Examination of **TR101** has been held on **13 October 2026** and accepted.

Signature of Internal Examiner Signature of External Examiner

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Abstract

The **TR-101: Tech Foundations – Linux and Web Basics** training programme provided practical knowledge of fundamental concepts related to Linux, web development, and basic computer technologies. The training was conducted through practical exercises, assignments, and project-based learning to develop technical and problem-solving skills.

During the training, two projects were developed. The first project, **"India Unfiltered: An Interactive Tourism Guide Website,"** was developed using HTML, CSS, and JavaScript. The website provides information about 20 popular tourist destinations in India through descriptions, images, videos, important facts, and Google Maps. The project demonstrates the use of basic web development technologies for creating an organized and responsive webpage.

The second project, **"File Integrity Checker Using Bash Shell Scripting on Linux,"** was developed using Bash Shell Scripting and Linux utilities. The project uses SHA-256 hash values to monitor files in a selected directory and detect modifications, deletions, and newly added files. It demonstrates the practical application of Linux commands, shell scripting, file management, and basic cybersecurity concepts.

The training helped in developing practical knowledge of HTML, CSS, JavaScript, Linux commands, Bash scripting, file handling, problem-solving, and technical documentation. Overall, the projects provided hands-on experience and established a foundation for further learning in web development, Linux administration, cybersecurity, and software development.

Acknowledgement

We would like to express my sincere gratitude to the Department of Information Technology, Guru Nanak Dev Engineering College, Ludhiana, for providing me with the opportunity to undertake the TR-101: *Tech Foundations – Linux and Web Basics* training during June–July 2026. The training provided valuable practical exposure to Linux, web development, and basic computer technologies.

We are thankful to the faculty members and training coordinators of the Department of Information Technology for their guidance, support, and encouragement throughout the training. Their guidance helped us understand the concepts of HTML, CSS, JavaScript, Linux commands, Bash scripting, and file management in a practical manner.

The training involved two projects: *India Unfiltered: An Interactive Tourism Guide Website* and *File Integrity Checker Using Bash Shell Scripting on Linux*, which provided valuable hands-on experience in web development, Linux, scripting, and basic cybersecurity.

Finally, We are grateful to everyone who directly or indirectly supported us in completing this training and preparing this report. The knowledge and practical experience gained during this training will serve as a useful foundation for our future academic and technical learning.

About the Institute

Guru Nanak Dev Engineering College (GNDEC), Ludhiana, is an engineering institution established in 1956 under the Nankana Sahib Education Trust (NSET). The foundation stone of the college was laid on 8 April 1956 by Dr. Rajendra Prasad, the first President of India. The institute was established with the aim of promoting technical education and supporting the social and economic development of society, particularly rural communities.

GNDEC offers undergraduate and postgraduate programmes in engineering and other professional fields. The college provides students with opportunities to develop technical knowledge and practical skills through academic learning, laboratory work, and project-based activities.

The Department of Information Technology focuses on developing students' computing knowledge, technical abilities, problem-solving skills, and professional competence. The TR-101 training programme is part of this learning process, providing practical exposure to Linux, web development, and basic computing technologies.

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Definitions, Acronyms and Abbreviations

Term	Definition
TR-101	Tech Foundations – Linux and Web Basics training course.
GNDEC	Guru Nanak Dev Engineering College.
IT	Information Technology.
B.Tech	Bachelor of Technology.
HTML	HyperText Markup Language; used to create and structure webpages.
CSS	Cascading Style Sheets; used to control the appearance and layout of webpages.
JavaScript	A programming language used to add interactive features to webpages.
Linux	A free and open-source operating system widely used for software development, servers, networking, and other computing environments.
Bash	A command-line shell used to execute Linux commands and automate tasks through shell scripts.
SHA-256	A cryptographic hash algorithm used in the project to generate hash values for checking file integrity.
GUI	Graphical User Interface; an interface that allows users to interact with a computer through graphical elements.
CLI	Command-Line Interface; an interface where users interact with a computer by entering commands.
W3C	World Wide Web Consortium, an organization responsible for developing web standards.
UGC	University Grants Commission.
VS Code	Visual Studio Code, a source-code editor used for writing and editing project files.
CSS3	The version of CSS used for modern webpage styling and responsive design.
HTML5	The modern version of HTML used to structure webpages and support multimedia and semantic elements.

CHAPTER 1
INTRODUCTION

1.1 About TR-101 Training

TR-101: Tech Foundations – Linux and Web Basics is a four-week foundation training program organized by the Department of Information Technology, Guru Nanak Dev Engineering College (GNDEC), Ludhiana. The training is specially designed for first-year B.Tech students to develop basic computing knowledge and practical skills required in higher semesters.

The primary objective of this training is to introduce students to Linux Operating System and Web Development technologies. During the training, students perform various laboratory exercises, assignments, and projects that help them understand both theoretical concepts and practical implementation.

The Linux module familiarizes students with the command-line environment, file system organization, user management, permissions, shell commands, and shell scripting. These skills enable students to efficiently interact with Linux-based systems that are widely used in software development and server administration.

The Web Development module introduces students to HTML and CSS, which are the fundamental technologies used for designing web pages. Students learn how to create static web pages, organize web content using HTML elements, and improve the appearance of websites using CSS.

Apart from technical knowledge, the training also helps students improve their logical thinking, problem-solving ability, teamwork, documentation skills, and confidence in using computers effectively.

Objectives of TR-101 Training

The major objectives of the training are:

1. To understand Linux fundamentals and use Linux commands for file management, directory navigation, and permissions.
2. To develop basic web pages using HTML and CSS, and implement simple interactive features using JavaScript.
3. To gain practical experience in Linux shell scripting and web development while improving problem-solving skills.

1.2 Overview of HTML

HTML stands for HyperText Markup Language. It is the standard markup language used to create and structure webpages. HTML provides the basic structure of a webpage and defines how different types of content such as headings, paragraphs, images, links, lists, forms, and other elements are organized.

HTML is not a programming language because it does not perform calculations or contain programming logic in the same way as languages such as C++ or Python. Instead, HTML uses predefined **tags and elements** to describe the structure and content of a webpage.

HTML5 is the modern version of HTML and provides support for multimedia and semantic webpage elements. It allows developers to include elements such as audio, video, images, forms, and other structured content without requiring additional technologies.

In the India Unfiltered project, HTML was used to create the main structure of the tourism website. Different sections were created for the introduction, destinations, images, videos, maps, navigation, and other information.

Features of HTML

The important features of HTML include:

- Simple and easy-to-understand syntax.
- Used to create the structure of webpages.
- Supports headings, paragraphs, lists, tables, links, and images.
- Supports multimedia elements such as audio and video.
- Provides semantic elements for better webpage organization.
- Supports forms for collecting user information.
- Works with CSS for webpage styling.
- Works with JavaScript for adding interactivity.
- Supported by all major modern web browsers.

Basic Structure of an HTML Document

A basic HTML document contains elements such as the document declaration, <html>, <head>, and <body>.

```
<!DOCTYPE html>

<html>

<head>

   <title>My Webpage</title>

</head>

<body>
```

```
    <h1>Welcome</h1>

    <p>This is a webpage.</p>

</body>

</html>
```

The <head> section contains information about the webpage, such as the title and links to stylesheets. The <body> section contains the visible content displayed in the browser.

HTML Tags

HTML uses tags to define different types of content. Some commonly used tags are:

- <h1> to <h6> – headings
- <p> – paragraph
- <a> – hyperlink
- <img> – image
- and <ol> – unordered and ordered lists
- – list item
- – general content section
- <section> – webpage section
- <nav> – navigation section
- <button> – button
- <video> – video content

Applications of HTML

HTML is used in:

- Website development
- Webpage design
- Online documentation
- Web forms
- Blogs and articles
- Educational websites
- E-commerce websites
- Tourism and information websites
- Web-based applications

Advantages of HTML

- Easy to learn and use.
- Free and widely supported.
- Works on different operating systems.
- Supported by all major web browsers.
- Can be combined with CSS and JavaScript.
- Provides a standard structure for webpages.

1.3 Overview of CSS

CSS stands for Cascading Style Sheets. It is a stylesheet language used to control the appearance and layout of HTML webpages. While HTML defines the structure and content of a webpage, CSS determines how the content is presented to the user.

CSS can be used to control colours, fonts, spacing, alignment, borders, backgrounds, images, buttons, page layouts, and responsive designs. It helps developers create webpages that are attractive, organized, readable, and user-friendly.

In the India Unfiltered project, CSS was used to style the webpage and create a responsive layout. It was used for fonts, colours, spacing, images, buttons, navigation elements, and different destination sections. CSS media queries were also used to make the website suitable for different screen sizes.

Types of CSS

There are three common ways to apply CSS to an HTML document.

1. Inline CSS

Inline CSS is applied directly to an HTML element using the style attribute.

** Welcome **

2. Internal CSS

Internal CSS is written inside the <style> element within the <head> section of an HTML document.

```
<style>
p {
   color: blue;
}
</style>
```

3. External CSS

External CSS is written in a separate .css file and connected to the HTML document.

```
<link rel="stylesheet" href="style.css">
```

External CSS is commonly preferred for larger websites because it keeps the structure and styling separate.

Example of CSS

```
body {
   font-family: Arial, sans-serif;
   margin: 0;
}

h1 {
   text-align: center;
}
```

Features of CSS

- Controls webpage appearance and layout.
- Provides control over colours and fonts.
- Controls spacing, margins, and padding.
- Supports responsive webpage design.
- Allows reusable styling rules.
- Supports backgrounds, borders, and effects.
- Separates webpage content from presentation.
- Works together with HTML and JavaScript.

Advantages of CSS

- Makes webpages more attractive.
- Provides consistent styling across multiple webpages.

- Reduces repetition in webpage design.
- Makes websites easier to maintain.
- Supports responsive design for different screen sizes.
- Improves the overall presentation and usability of websites.

Applications of CSS

CSS is commonly used in:

- Website design
- Responsive web development
- Navigation menus
- Forms and buttons
- Webpage layouts
- Portfolio websites
- E-commerce websites
- Tourism and information websites
- Interactive web interfaces

1.4 Overview of Linux

Linux is a **free and open-source operating system** based on the Unix operating system. It was developed by **Linus Torvalds in 1991** and is widely used in servers, software development, networking, cloud computing, embedded systems, and other computing environments.

Linux provides both graphical and command-line interfaces. The **Terminal** allows users to interact directly with the operating system by entering commands. This makes it possible to perform operations such as creating and managing files, navigating directories, changing permissions, searching for information, and executing shell scripts.

During the TR-101 training, Linux was used to understand file management, directory

operations, permissions, command-line utilities, and shell scripting.

The Linux project developed during the training, **File Integrity Checker Using Bash Shell Scripting on Linux**, demonstrates the practical use of the Linux command-line environment. The project uses Bash Shell Scripting and Linux utilities to generate and compare SHA-256 hash values for files.

Features of Linux

The important features of Linux include:

- Free and open-source operating system.
- Supports command-line and graphical interfaces.
- Provides strong file and user permission management.
- Supports multiple users.
- Supports multitasking.
- Provides a large number of command-line utilities.
- Supports shell scripting and automation.
- Provides good security and stability.
- Available in different distributions such as Ubuntu and Fedora.

Linux File System

Linux organizes files and directories in a hierarchical file system. The root directory is represented by /. Important directories include:

- /home – contains users' personal files.
- /etc – contains system configuration files.
- /bin – contains commonly used executable commands.
- /var – contains variable data such as logs.
- /tmp – contains temporary files.
- /root – home directory of the root user.

The file system structure allows users and applications to organize and access data efficiently.

Common Linux Commands

Some commonly used Linux commands are:

Command	Purpose
pwd	Displays the current working directory
ls	Lists files and directories
cd	Changes the current directory
mkdir	Creates a directory
touch	Creates a file
cp	Copies files or directories
mv	Moves or renames files
rm	Removes files or directories
cat	Displays file contents
grep	Searches for specific text
chmod	Changes file permissions
find	Searches for files and directories
sha256sum	Generates SHA-256 hash values

Advantages of Linux

- Free and open-source.
- Stable and reliable.
- Provides powerful command-line tools.
- Supports automation through shell scripting.
- Offers strong permission and security mechanisms.
- Suitable for software development and server environments.
- Can run efficiently on different hardware configurations.

- Provides extensive community and documentation support.

Applications of Linux

Linux is widely used in:

- Web servers
- Cloud computing
- Software development
- Network administration
- Cybersecurity
- Database servers
- Embedded systems
- Supercomputers
- DevOps and automation
- System administration

1.5 Software and Tools Used

The following software, programming languages, development tools, and utilities were used during the TR-101 training and for the development of the projects.

Software and Development Tools

Software / Tool	**Purpose**
Visual Studio Code	Used for writing and editing HTML, CSS, JavaScript, and Bash files
Google Chrome	Used for testing and viewing the tourism website
Microsoft Edge	Used for additional browser testing
Ubuntu Linux	Used for Linux commands and Bash shell scripting

Bash Shell	Used to develop the File Integrity Checker
Git / GitHub	Used for optional version control and project management
Google Maps	Used to display locations of tourist destinations
YouTube	Used for embedding tourism-related videos

Programming and Web Technologies

HTML5: Used to create the structure and content of the tourism website.

CSS3: Used to design the appearance, layout, and responsive behaviour of the website.

JavaScript: Used to provide simple interactive features such as navigation, smooth scrolling, and other user interactions.

Bash Shell Scripting: Used to implement the File Integrity Checker and automate Linux-based file monitoring tasks.

Linux Utilities

The following Linux utilities were used in the File Integrity Checker project:

- find
- sha256sum
- grep
- awk
- cp
- mkdir
- date
- chmod

These utilities were combined with Bash scripting to perform file searching, hashing, text processing, directory operations, permission management, and logging.

Hardware Requirements

The projects can be developed and executed on a basic computer system with the following minimum requirements:

- **Processor:** Intel Core i3 or above or equivalent AMD processor.
- **RAM:** Minimum 4 GB; 8 GB recommended.
- **Storage:** At least 500 MB of free storage.
- **Display:** Minimum resolution of 1366 × 768.
- **Operating System:** Windows, Linux, or macOS for the web project; Ubuntu Linux or another compatible Linux distribution for the Linux project.
- **Internet Connection:** Required for external resources such as online images, YouTube videos, and Google Maps.

CHAPTER 2

HTML/CSS PROJECT

2.1 Introduction

The project developed during the TR-101 training is a tourism website named “India Unfiltered: An Interactive Tourism Guide Website”. The project was developed to provide organized information about popular tourist destinations in India through a single responsive web page. The website covers 20 tourist destinations and presents their descriptions, important facts, images, videos, and map locations.

The website was developed using HTML, CSS, and JavaScript. HTML was used to create the structure and organize the content, CSS was used for styling and responsive layout, and JavaScript was used for simple interactive features such as navigation, smooth scrolling, and returning to the top of the page. The project demonstrates the practical application of basic web development concepts learned during the training.

2.2 Problem Statement

Information about tourist places in India is available on many different websites. Users often need to visit multiple sources to collect basic information, images, videos, and location details before planning a visit. This makes the process less convenient and less organized.

The purpose of this project was to create one simple and user-friendly website where users can explore information about different tourist destinations in India. The website brings destination descriptions, images, videos, quick facts, and map locations together in one place.

2.3 Requirement Analysis

The project requires a computer or mobile device with a modern web browser. An internet connection is required for loading external videos, Google Maps, and other external resources. The website was developed using HTML5, CSS3, and JavaScript, with Visual Studio Code used as the code editor.

Table 2.1 Software and Hardware Requirements

Requirement	Details
Operating System	Windows 10/11, Linux, or macOS
Languages	HTML5, CSS3, JavaScript
Code Editor	Visual Studio Code
Web Browser	Google Chrome, Microsoft Edge, or Mozilla Firefox
Version Control	Git and GitHub (optional)
Processor	Intel Core i3 or above, or equivalent
RAM	Minimum 4 GB; 8 GB recommended
Storage	At least 500 MB free space
Display	Minimum 1366 × 768 resolution
Internet	Required for external videos, maps, and resources

2.4 Design and Layout

The website was designed as a single responsive webpage. It contains a header, an introduction section, navigation, destination sections, images, videos, and maps. The homepage uses the heading “20 Places. One India. No Filter.” to introduce the purpose of the website.

A navigation section allows users to move between different destination sections. The website contains information about Taj Mahal, Jaipur, Goa, Kerala Backwaters, Leh-Ladakh, Varanasi, Golden Temple, Mysore Palace, Hampi, Rann of Kutch, Darjeeling, Shimla, Manali, Ooty, Andaman and Nicobar Islands, Khajuraho Temples, Konark Sun Temple, Sikkim, Udaipur, and Charminar.

The layout is responsive, so the arrangement of content changes according to screen size. This allows the website to be viewed on desktop computers, tablets, and mobile devices.

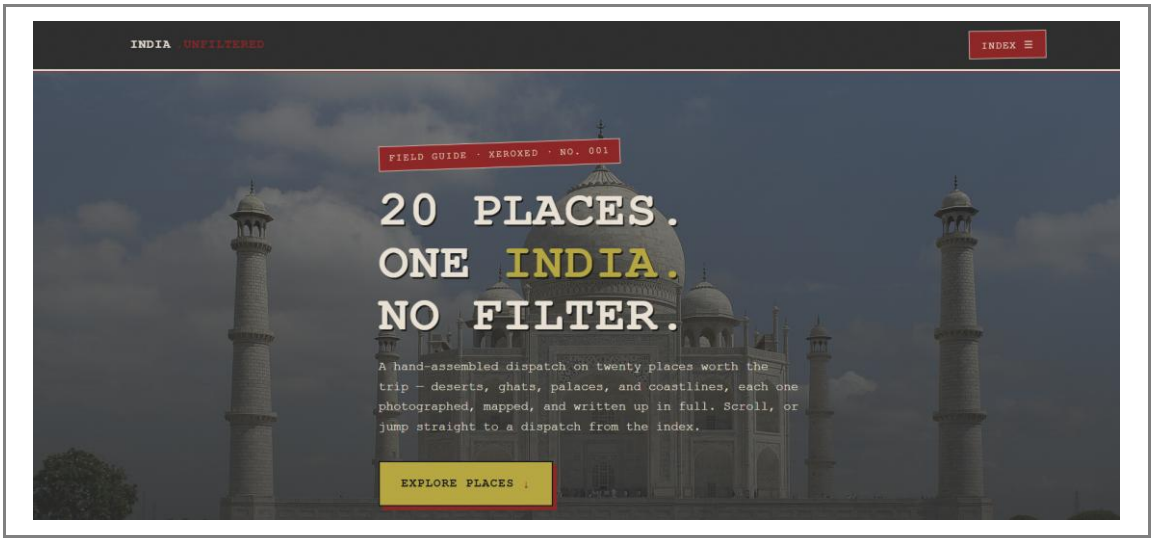


Figure 2.1 Overall layout of the India Unfiltered tourism website

2.5 HTML Structure

HTML was used to create the basic structure of the website. Different HTML elements were used to organize headings, paragraphs, images, links, buttons, navigation, and destination sections. The content was divided into meaningful sections so that users can easily read and navigate through the website.

The HTML structure also provides locations for multimedia content. Images are placed with the relevant destinations, while embedded YouTube videos and Google Maps are included to provide additional information and location details.



Figure 2.2 HTML structure of the tourism website

2.6 CSS Styling

CSS was used to control the appearance and layout of the website. It was applied to fonts, colours, spacing, images, buttons, navigation elements, and page sections. CSS helped in creating a consistent and readable design throughout the webpage.

CSS media queries were used to make the website responsive. This allows the webpage layout to adjust according to the screen size and improves usability on computers, tablets, and mobile devices. JavaScript was also used for simple interactive features such as opening and closing the destination menu, smooth scrolling, and returning to the top of the page.

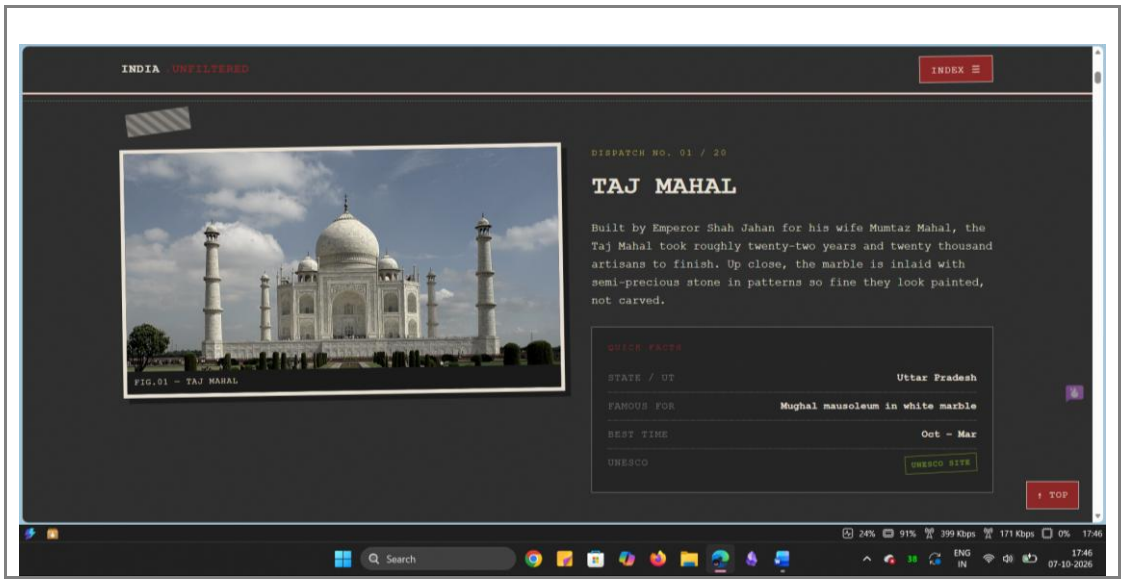


Figure 2.3 Responsive design and CSS styling

2.7 Screenshots

Screenshots of the completed website should be included in this section to show the practical implementation of the project. The screenshots should be captured from the actual project and should clearly show the webpage, destination sections, multimedia content, and responsive views.

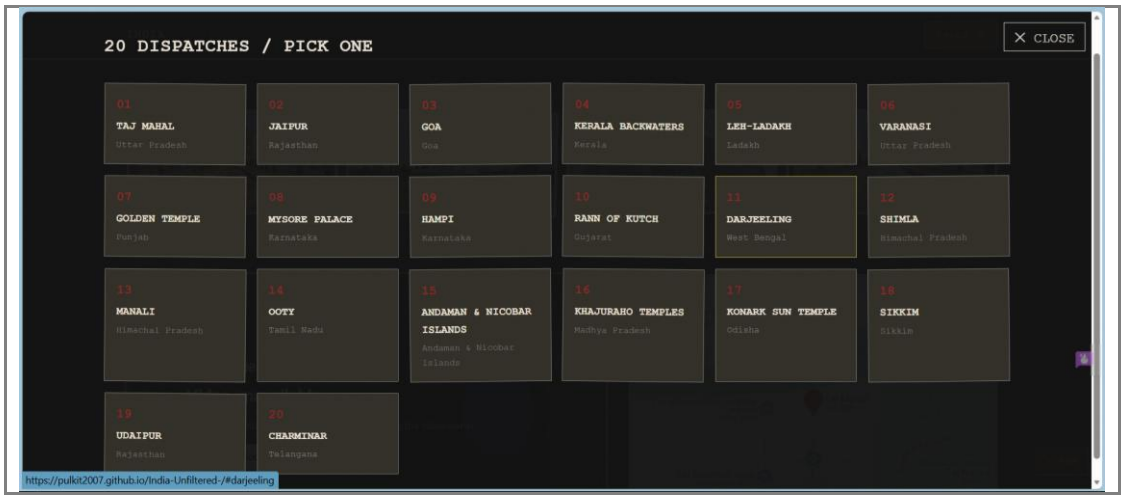


Figure 2.4 Homepage of the India Unfiltered website

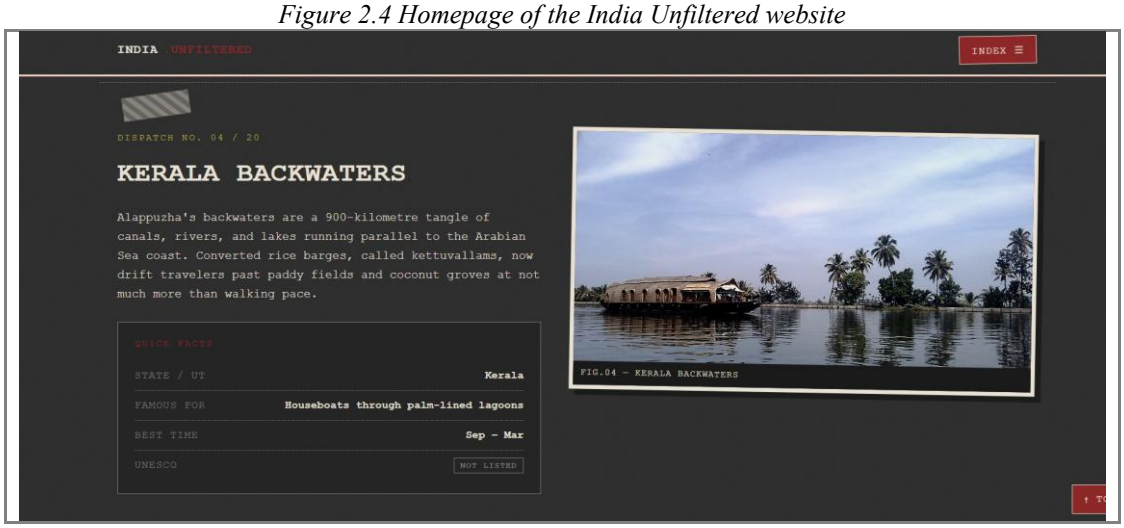


Figure 2.5 Destination section showing text and images

2.8 Learning Outcomes

Through this project, the fundamentals of HTML, CSS, and JavaScript were applied in a practical web development task. The project helped in understanding how webpage sections are created, how content is organized, and how CSS is used to improve the appearance and responsiveness of a website.

Created structured webpage sections using HTML.

Used CSS for webpage styling, spacing, layout, and responsiveness.

Added images, embedded videos, and Google Maps to a webpage.

Created navigation and simple interactive features using JavaScript.

Tested the website in a web browser and checked its layout on different screen sizes.

CHAPTER 3

LINUX PROJECT

3.1 Introduction

The Linux project developed during the training is titled “File Integrity Checker Using Bash Shell Scripting on Linux”. The project is designed to monitor files in a selected directory and identify changes that may affect their integrity.

The project uses Bash Shell Scripting and the SHA-256 hashing algorithm. A baseline set of hash values is generated for the files in the selected directory. During verification, the current hash values are compared with the stored values. This makes it possible to identify modified, deleted, or newly added files.

The project demonstrates the practical use of Linux commands, shell scripting, file handling, hashing, and basic cybersecurity concepts in a simple command-line environment.

3.2 Problem Statement

Modern computer systems store a large amount of important and sensitive information. Files may be modified, deleted, or replaced intentionally by unauthorized users or accidentally because of human error. When a directory contains many files, identifying such changes manually is difficult.

Existing file monitoring tools may require additional software or advanced configuration. Therefore, the project aims to provide a simple, lightweight solution that can verify the integrity of files in a selected directory by comparing SHA-256 hash values.

3.3 Linux Commands Used

The following Linux commands and utilities are relevant to the implementation of the File Integrity Checker. They are used for directory navigation, file management, hashing, searching, text processing, permissions, and report generation.

Table 3.1 Linux Commands and Utilities Used

Command / Utility	Purpose
find	Searches for files in a directory and its subdirectories.
sha256sum	Generates and verifies SHA-256 hash values.
grep	Searches for matching text or records.
awk	Processes and extracts text data.
cp	Copies files.
mkdir	Creates directories.
date	Provides date and time information for logs.
chmod	Changes file permissions.

3.4 Shell Commands and Utilities

Bash is a command-line shell that allows users to execute Linux commands and automate repetitive tasks through shell scripts. In this project, Bash provides the logic required to generate file hashes, store baseline information, compare current values, and report changes.

Variables are used to store directory paths, file names, hash values, and other information.

Commands such as find and sha256sum are used to identify files and generate their hashes.

grep and awk are used for searching and processing stored integrity information.

cp and mkdir are used for file and directory operations.

date is used for recording information in logs.

chmod is used to provide the required execution permissions to the shell script.

Conditional statements and loops can be used to compare results and identify file changes.

3.5 Implementation

The implementation of the File Integrity Checker follows a simple sequence. First, a directory to be monitored is selected. The script scans the directory and generates SHA-256 hash values for the files. These values are stored as the baseline integrity database. During a later verification, the script generates the current hash values and compares them with the stored values.

Step 1: Open the Linux Terminal and select the directory to be monitored.

Step 2: Create the project directory using mkdir.

Step 3: Use find to locate the files inside the selected directory.

Step 4: Generate SHA-256 hashes using sha256sum.

Step 5: Store the generated hashes as the integrity baseline.

Step 6: Run the verification process and generate the current hashes.

Step 7: Compare current hashes with the stored baseline.

Step 8: Identify modified, deleted, or newly added files.

Step 9: Record the detected changes in logs or reports.

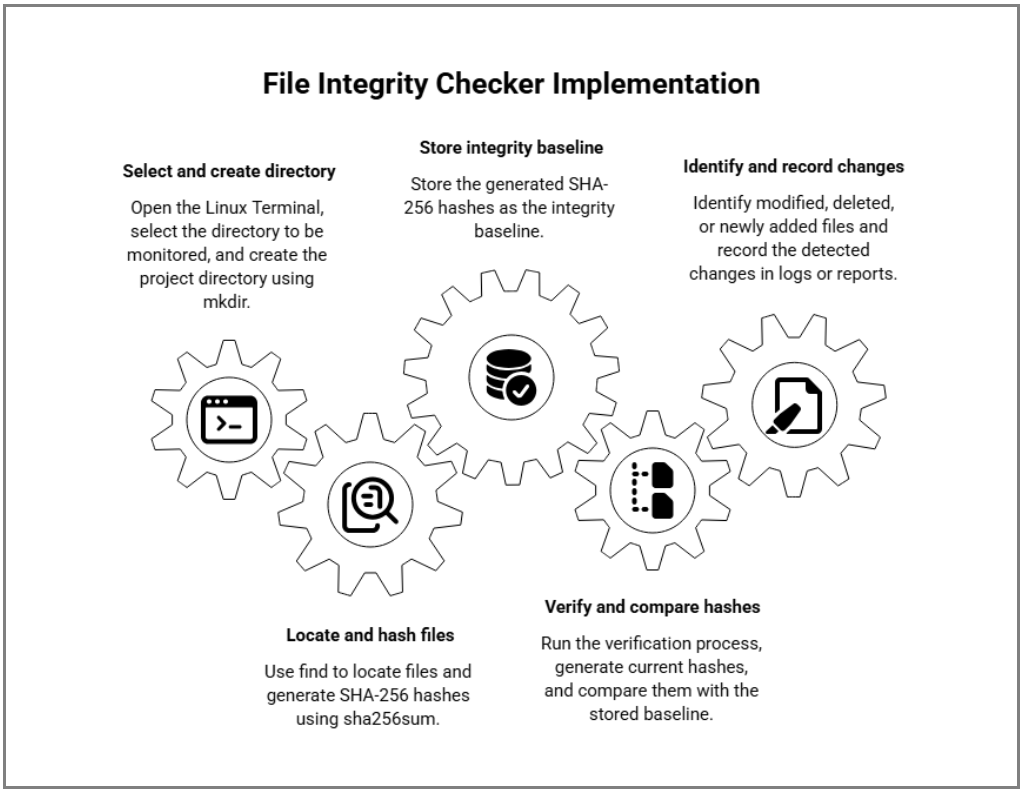


Figure 3.1 File integrity monitoring process

The implementation is intended for local file monitoring through the Linux command line. The project does not provide real-time monitoring, cloud support, or a graphical user interface in its current form.

3.6 Terminal Output Screenshots

Screenshots of the Linux terminal should be included to demonstrate the execution of the project. The screenshots should be taken from the actual implementation and should show the commands and outputs clearly.

```
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ mkdir test
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ echo "Hello World" > test/f
ile1.txt
echo "Linux Project" > test/file2.txt
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ ls test
file1.txt  file2.txt
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ ./file_integrity_checker.sh
 init test
Creating baseline for: test
Baseline created successfully.
Saved as: baseline.sha256
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$
```

Figure 3.2 Creating the project and monitored directory

```
    # Check for new files
    find "$DIRECTORY" -type f ! -name "$BASELINE" | while read -r file
    do
        if ! grep -Fq "  $file" "$BASELINE"; then
            echo "[NEW]      $file"
        fi
    done

    echo
    echo "Integrity check completed."
}
```

Figure 3.4 Creating or storing the integrity baseline

```
check_integrity() {
    if [ ! -f "$BASELINE" ]; then
        echo "Error: Baseline file not found."
        echo "Run: $0 init <directory>"
        exit 1
    fi

    echo "Checking file integrity..."
    echo
```

Figure 3.5 Verifying files against the stored hashes

```
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ ./file_integrity_checker.sh init test
Creating baseline for: test
Baseline created successfully.
Saved as: baseline.sha256
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ echo "Modified content" > test/file1.txt
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ ./file_integrity_checker.sh check test
Checking file integrity...

[MODIFIED] test/file1.txt

Integrity check completed.
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ rm test/file2.txt
echo "New file" > test/file3.txt
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$ ./file_integrity_checker.sh check test
Checking file integrity...

[MODIFIED] test/file1.txt
[DELETED]  test/file2.txt
[NEW]      test/file3.txt

Integrity check completed.
lenovo@Pulkit:/mnt/c/Users/LENOVO/file-integrity-checker1$
```

Figure 3.6 Detecting a modified, deleted, or newly added file

3.7 Learning Outcomes

After completing the Linux project, the practical use of Linux commands and Bash shell scripting was understood. The project also introduced the use of SHA-256 hashing for checking file integrity and demonstrated how command-line tools can be combined to solve a practical problem.

Understood the Linux terminal environment and command-line workflow.

Used Linux utilities for file and directory operations.

Learned the practical use of Bash shell scripting.

Generated and compared SHA-256 hash values.

Understood how file modifications, deletions, and additions can be detected.

Learned basic concepts related to file integrity and cybersecurity.

Improved problem-solving and command-line troubleshooting skills.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 HTML Project Results

The India Unfiltered tourism website was successfully developed as a responsive web project. The website provides information about 20 tourist destinations in India and combines text, images, videos, quick facts, and Google Maps in a single platform.

The use of HTML provided a structured arrangement of the webpage content, while CSS improved the visual presentation, spacing, navigation, and responsiveness. JavaScript was used for simple interactive features. The project demonstrated that basic front-end technologies can be combined to create an organized and user-friendly tourism information website.

4.2 Linux Project Results

The File Integrity Checker was designed as a lightweight Linux-based solution for checking the integrity of files in a selected directory. SHA-256 hash values are used as the basis for comparison between the stored baseline and the current state of the files.

The project can identify changes such as file modification, deletion, and addition. Linux utilities including find, sha256sum, grep, awk, cp, mkdir, date, and chmod support the implementation. The project demonstrates how Bash scripting and standard Linux utilities can be combined to automate a basic file security task.

4.3 Comparison of Learning Outcomes

Table 4.1 Comparison of Learning Outcomes

HTML/CSS Project	Linux Project
Learned webpage structure and organization	Learned Linux command-line operations
Applied CSS for layout and presentation	Used Bash shell scripting
Created responsive webpage sections	Managed files and directories
Integrated images, videos, and maps	Generated and compared SHA-256 hashes
Implemented basic JavaScript interaction	Detected file integrity changes
Tested a website in a browser	Used Linux utilities for automation

Both projects provided practical experience with technologies introduced during TR-101. The HTML/CSS project focused mainly on web development and user interface design, whereas the Linux project focused on command-line operations, shell scripting, file management, and basic security concepts.

4.4 Skills Acquired

HTML programming and webpage structure

CSS styling and responsive web design

Basic JavaScript interaction

Multimedia integration in webpages

Linux command-line operations

Bash shell scripting

File and directory management

SHA-256 hashing and file integrity checking

Problem-solving and logical thinking

Technical documentation and project presentation

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The TR-101 training provided practical knowledge of Linux and web development through hands-on exercises and project work. The HTML/CSS project, India Unfiltered, demonstrated how HTML, CSS, and JavaScript can be used to organize information and create a responsive tourism website with multimedia content.

The Linux project, File Integrity Checker Using Bash Shell Scripting on Linux, demonstrated how Linux commands and Bash scripting can be used to perform a practical file monitoring task. The use of SHA-256 hash comparison provided a simple method for detecting file modifications, deletions, and newly added files.

Together, the projects improved practical programming, problem-solving, documentation, command-line, and web development skills. The training provided a foundation for further learning in areas such as advanced JavaScript, full-stack web development, Linux administration, cybersecurity, databases, and cloud computing.

5.2 Future Scope

In the future, the website can be improved by adding more tourist destinations and providing updated information about them. A search feature can be added to help users find places easily. The website can also be made more mobile-friendly and include useful details such as nearby attractions, travel tips, and the best time to visit.

In the future, the file integrity checker can be improved by adding automatic scheduled scans and saving the results in log files. Email or desktop notifications can be added to alert users when files are modified, deleted, or newly added. The tool can also be extended to monitor multiple directories.

These enhancements can make both projects more useful for real-world applications while providing further opportunities to develop skills in web development, automation, system administration, and cybersecurity.

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